

Practice Problems (Data Types)

Required knowledge

Operators, Data types, Variables and expression, Basic input/output

1. **Print Hello World:** Write a program to print "Hello, World!" to the console.
2. **Print Your Name:** Prompt the user to enter their name and print "Hello, [Name]!"
3. **Display Data Types:** Declare variables of different data types (e.g., int, float, char) and display them.
4. **Simple Arithmetic:** Take two integers as input and print their sum, difference, product, and quotient.
5. **User Input Variables:** Prompt the user to enter values for different data types (int, float, char, string) and display them.
6. **Area of a Rectangle:** Take length and width as inputs, calculate the area of the rectangle, and display it.
7. **Temperature Converter:** Convert a temperature from Celsius to Fahrenheit. Formula: $F = (C \times 9/5) + 32$.
8. **Circle Calculations:** Take the radius of a circle as input, calculate and print the area and circumference. Use $\pi=3.14159$.
9. **Basic Calculator:** Prompt the user for two numbers and an operator (+, -, *, /). Perform the corresponding operation and print the result.
10. **Average Calculator:** Take three integers as input and calculate their average.
11. **Modulus Operation:** Take two integers as input and print the remainder when the first is divided by the second.
12. **Swap Two Variables:** Write a program to swap the values of two variables without using a third variable.
13. **Square and Cube:** Prompt the user for an integer and print its square and cube.
14. **Type Casting:** Take a float input and display its integer part by converting it explicitly.
15. **ASCII Value:** Take a character input and print its ASCII value.

16. **Quadratic Equation Solver:** Take values for a, b, and c and calculate the discriminant of the equation $ax^2 + bx + c = 0$. Print the discriminant only.

17. **Time Conversion:** Convert a total number of seconds entered by the user into hours, minutes, and seconds.

18. **Compound Interest:** Calculate compound interest. Formula:

$A = P(1 + \frac{r}{n})^{n \times t}$, where P is the principal amount, r is the rate, n is the number of times interest is compounded per year, and t is the time in years.

19. **Distance Calculator:** Given the coordinates of two points, calculate the distance between them. Formula: $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$.

20. **BMI Calculator:** Take a person's weight (kg) and height (m), and calculate the Body Mass Index (BMI). Formula: $BMI = \frac{weight}{height^2}$.

21. **Digit Sum:** Prompt the user for a 3-digit integer and calculate the sum of its digits.

22. **Reverse a Number:** Take a 3-digit integer as input and print it in reverse.

23. **Currency Denominations:** Given an amount in dollars, calculate the minimum number of \$100, \$50, \$20, \$10, \$5, and \$1 bills needed to make that amount.

24. **Day Conversion:** Given a number of days, convert it into weeks and days (e.g., 10 days = 1 week and 3 days).

25. **Character Manipulation:** Take a lowercase character input and print its uppercase equivalent (without using conditionals, assume it's lowercase).

26. **Factorial Calculation (Approximation):** Take a non-negative integer and approximate its factorial using Stirling's approximation formula.

27. **Sum of Arithmetic Series:** Take inputs for the first term, common difference, and number of terms of an arithmetic series, and calculate the sum using the formula $S = \frac{n}{2}(2a + (n - 1)d)$.

28. **Distance Conversion:** Given a distance in kilometers, convert it to miles. (1 kilometer = 0.621371 miles)

- 29.**Average of N Numbers:** Prompt the user to input how many numbers they want to average, take those numbers as input, and calculate the average.
- 30.**Floating Point Precision:** Take a floating-point number and print it with 2, 3, and 5 decimal places.
- 31.**Monthly Salary Calculator:** Prompt for basic salary, calculate HRA (20%), DA (50%), and display the gross salary (basic + HRA + DA).
- 32.**Random Number Properties:** Generate a random number within a specified range and display it.
- 33.**Simple Product Inventory:** Given the cost of an item and the number of items in stock, calculate the total inventory value.
- 34.**Date Converter:** Given a date in the format YYYYMMDD, print it as "Year: YYYY, Month: MM, Day: DD".
- 35.**Investment Growth:** Given an initial investment, annual interest rate, and number of years, calculate the future investment value using the formula $FV = PV \times \left(1 + \frac{rate}{100}\right)^{years}$.